

Finite-Blocklength Latency Minimization for Uplink and Downlink Communications: Topic-Split Summary Tables

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1 Naming

Table 1: Document naming

Document label	Meaning
Training Only Method	Direct online optimization during scheduling.
Training + Testing Method	Separate training stage followed by testing-stage inference.
Payload completion	Each user continues until its remaining payload is exhausted.

2 Coverage

2.1 Uplink

Table 2: Uplink coverage used in this document

Scenario	Training Only	Training + Testing	Config
Payload completion	Seed 3	Train [1, 2], test 3	UL payload
Payload completion direction and weighting sweep	Seed 3, unweighted / inverse-CNR, asc. / desc.	Train [1, 2], test 3	Sweep 20260727_215427
Payload completion binary test-time search	–	Shared trained model, test 3, asc. / desc. / binary	Held-out search

2.2 Downlink

Table 3: Downlink coverage used in this document

Scenario	Training Only	Training + Testing	Config
Payload completion	Seed 3	Train [1, 2], test 3	DL payload
Payload completion direction and weighting sweep	Seed 3, unweighted / inverse-CNR, asc. / desc.	Train [1, 2], test 3	Sweep 20260727_215427

3 Latency, Synchronization, and Service

3.1 Uplink

Table 4: Uplink latency, synchronization, and service summary

Metric	Payload completion	
	Train Only	Train+ Test
Init. total	0.030733	0.030733
Final total	0.024667	0.025200
Red. (%)	19.7397	18.0043
Init. avg	0.010244	0.010244
Final avg	0.008222	0.008400
Init. async.	0.016667	0.016667
Final async.	0.015333	0.015333
Async. red. (%)	8.0000	8.0000
Served bits	6000	6000
Skipped blocks	0	0

3.2 Downlink

Table 5: Downlink latency, synchronization, and service summary

Metric	Payload completion	
	Train Only	Train+ Test
Init. total	0.305000	0.305000
Final total	0.035800	0.073533
Red. (%)	88.2623	75.8907
Init. avg	0.101667	0.101667
Final avg	0.011933	0.024511
Init. async.	0.131600	0.131600
Final async.	0.007867	0.046400
Async. red. (%)	94.0223	64.7416
Served bits	6000	6000
Skipped blocks	3	10

4 Radio Quality

4.1 Uplink

Table 6: Uplink radio-quality summary

Metric	Payload completion	
	Train Only	Train+ Test
Init. SNR (dB)	8.1078	8.1078
Final SNR (dB)	8.2627	8.3355
Init. SINR (dB)	-3.3939	-3.3939
Final SINR (dB)	-3.2868	-3.4146

4.2 Downlink

Table 7: Downlink radio-quality summary

Metric	Payload completion	
	Train Only	Train+ Test
Init. SNR (dB)	8.0000	8.0000
Final SNR (dB)	13.9912	9.9357

5 Training and Runtime

5.1 Uplink

Table 8: Uplink training and runtime summary

Metric	Payload completion	
	Train Only	Train+ Test
Train episodes	–	2
Epoch plan	–	max 100 kkt_converged, kkt_converged, kkt_converged
Solve status	–	kkt_converged
Feasible queries	–	87 / 92
Best mean rate	–	15.371869
Train time (s)	95.283	282.902
Test time (s)	0.000	52.887
Total time (s)	95.283	335.790
FB evals	835	7268
Fwd evals	97	284
Opt. steps	–	334
Opt. updates	738	–

5.2 Downlink

Table 9: Downlink training and runtime summary

Metric	Payload completion	
	Train Only	Train+ Test
Train episodes	–	2
Epoch plan	–	120 / 96
Solve status	–	kkt_converged
Feasible queries	–	26 / 51 (0.509804)
Best mean rate	–	17.961083
Best sum rate	–	22.545255
Train time (s)	419.069	311.368
Test time (s)	0.000	1.764
Total time (s)	419.069	313.132
FB evals	3525	0
Fwd evals	1238	118
Opt. steps / updates	2287	193

6 Ascending vs. Descending Search

This comparison isolates search order. The objective and scenario are kept fixed so the tables show whether changing the traversal direction changes the final schedule quality or mainly the runtime.

6.1 Uplink

Table 10: Uplink payload-completion comparison for ascending vs. descending search

Metric	Payload completion			
	Training Only Desc.	Training Only Asc.	Training + Testing Desc.	Training + Testing Asc.
Init. total	0.030733	0.030733	0.030733	0.030733
Final total	0.024667	0.024667	0.024933	0.027467
Red. (%)	19.7397	19.7397	18.8720	10.6291
Init. avg	0.010244	0.010244	0.010244	0.010244
Final avg	0.008222	0.008222	0.008311	0.009156
Init. async.	0.016667	0.016667	0.016667	0.016667
Final async.	0.015333	0.015333	0.015333	0.016267
Async. red. (%)	8.0000	8.0000	8.0000	2.4000
Init. SNR (dB)	8.1078	8.1078	8.1078	8.1078
Final SNR (dB)	8.4196	8.2859	8.3865	8.1852
Init. SINR (dB)	-3.3939	-3.3939	-3.3939	-3.3939
Final SINR (dB)	-3.3022	-3.3125	-3.3269	-3.4335
Train time (s)	34.315	118.882	146.092	240.023
Test time (s)	0.000	0.000	43.488	64.752
Total time (s)	34.315	118.882	189.581	304.776

6.2 Downlink

Table 11: Downlink payload-completion comparison for ascending vs. descending search

Metric	Payload completion			
	Training Only Desc.	Training Only Asc.	Training + Testing Desc.	Training + Testing Asc.
Init. total	0.305000	0.305000	0.305000	0.305000
Final total	0.049333	0.049333	0.075667	0.180467
Red. (%)	83.8251	83.8251	75.1913	40.8306
Init. avg	0.101667	0.101667	0.101667	0.101667
Final avg	0.016444	0.016444	0.025222	0.060156
Init. async.	0.131600	0.131600	0.131600	0.131600
Final async.	0.003733	0.003733	0.048267	0.066800
Async. red. (%)	97.1631	97.1631	63.3232	49.2401
Init. SNR (dB)	8.0000	8.0000	8.0000	8.0000
Final SNR (dB)	14.1792	14.1792	9.3594	7.5048
Init. SINR (dB)	-2.8373	-2.8373	-2.8373	-2.8373
Final SINR (dB)	8.2492	8.2492	11.9746	2.3859
Train time (s)	148.755	375.812	117.507	1190.145
Test time (s)	0.000	0.000	1.976	7.217
Total time (s)	148.755	375.812	119.483	1197.362

7 Binary Test-Time Search

This comparison keeps the Training + Testing method fixed and changes only the held-out test search rule. The goal is to check whether binary search reduces test-time effort without changing the final latency outcome.

7.1 Uplink

Table 12: Uplink Training + Testing payload-completion comparison for test-time search rules

Metric	Descending fixed-step	Ascending fixed-step	Binary
Init. total	0.030733	0.030733	0.030733
Final total	0.024933	0.024933	0.024933
Red. (%)	18.8720	18.8720	18.8720
Init. avg	0.010244	0.010244	0.010244
Final avg	0.008311	0.008311	0.008311
Init. async.	0.016667	0.016667	0.016667
Final async.	0.015333	0.015333	0.015333
Async. red. (%)	8.0000	8.0000	8.0000
Init. SNR (dB)	8.1078	8.1078	8.1078
Final SNR (dB)	8.3865	8.3865	8.3865
Init. SINR (dB)	-3.3939	-3.3939	-3.3939
Final SINR (dB)	-3.3269	-3.3269	-3.3269
Train time (s)	146.092	146.092	146.092
Test time (s)	82.205	81.313	81.076
Total time (s)	228.298	227.405	227.168
Inference forward calls	288	145	153
Test candidate $n_{k,\ell}$ states	154	11	19
Test-time red. vs desc. (%)	0.0000	1.0856	1.3739
Forward-call red. vs desc. (%)	0.0000	49.6528	46.8750
State-count red. vs desc. (%)	0.0000	92.8571	87.6623

7.2 Downlink

Table 13: Downlink Training + Testing payload-completion comparison for test-time search rules

Metric	Descending fixed-step	Ascending fixed-step	Binary
Init. total	0.305000	0.305000	0.305000
Final total	0.075667	0.075667	0.075667
Red. (%)	75.1913	75.1913	75.1913
Init. avg	0.101667	0.101667	0.101667
Final avg	0.025222	0.025222	0.025222
Init. async.	0.131600	0.131600	0.131600
Final async.	0.048267	0.048267	0.048267
Async. red. (%)	63.3232	63.3232	63.3232
Init. SNR (dB)	8.0000	8.0000	8.0000
Final SNR (dB)	9.3594	9.3594	9.3594
Init. SINR (dB)	-2.8373	-2.8373	-2.8373
Final SINR (dB)	11.9746	11.9746	11.9746
Train time (s)	117.507	117.507	117.507
Test time (s)	3.502	4.148	3.127
Total time (s)	121.010	121.655	120.634
Inference forward calls	86	163	47

Table 13: Downlink Training + Testing payload-completion comparison for test-time search rules (continued)

Metric	Descending fixed-step	Ascending fixed-step	Binary
Per-user test forward calls	25, 19, 42	48, 54, 61	17, 12, 18
Test-time red. vs desc. (%)	0.0000	-18.4320	10.7167
Forward-call red. vs desc. (%)	0.0000	-89.5349	45.3488

8 Inverse-CNR Weighting vs. Unweighted Objective

This comparison keeps the search direction fixed to descending and changes only the objective weighting. The tables show whether inverse-CNR weighting improves latency, synchronization, or radio quality relative to the unweighted objective.

8.1 Uplink

Table 14: Uplink payload-completion comparison for inverse-CNR weighting

Metric	Training Only Unweighted	Training Only Inv.-CNR	Training + Testing Unweighted	Training + Testing Inv.-CNR
Init. total	0.030733	0.030733	0.030733	0.030733
Final total	0.024667	0.024733	0.024933	0.024933
Red. (%)	19.7397	19.5228	18.8720	18.8720
Init. avg	0.010244	0.010244	0.010244	0.010244
Final avg	0.008222	0.008244	0.008311	0.008311
Init. async.	0.016667	0.016667	0.016667	0.016667
Final async.	0.015333	0.015733	0.015333	0.015333
Async. red. (%)	8.0000	5.6000	8.0000	8.0000
Init. SNR (dB)	8.1078	8.1078	8.1078	8.1078
Final SNR (dB)	8.4196	8.3587	8.3865	8.4089
Init. SINR (dB)	-3.3939	-3.3939	-3.3939	-3.3939
Final SINR (dB)	-3.3022	-3.3800	-3.3269	-3.3250
Train time (s)	34.315	35.132	146.092	135.961
Test time (s)	0.000	0.000	43.488	47.455
Total time (s)	34.315	35.132	189.581	183.416

8.2 Downlink

Table 15: Downlink payload-completion comparison for inverse-CNR weighting

Metric	Training Only Unweighted	Training Only Inv.-CNR	Training + Testing Unweighted	Training + Testing Inv.-CNR
Init. total	0.305000	0.305000	0.305000	0.305000
Final total	0.049333	0.040267	0.075667	0.073600
Red. (%)	83.8251	86.7978	75.1913	75.8689
Init. avg	0.101667	0.101667	0.101667	0.101667
Final avg	0.016444	0.013422	0.025222	0.024533
Init. async.	0.131600	0.131600	0.131600	0.131600
Final async.	0.003733	0.013467	0.048267	0.021733
Async. red. (%)	97.1631	89.7670	63.3232	83.4853
Init. SNR (dB)	8.0000	8.0000	8.0000	8.0000
Final SNR (dB)	14.1792	15.3203	9.3594	9.8556
Init. SINR (dB)	-2.8373	-2.8373	-2.8373	-2.8373

Table 15: Downlink payload-completion comparison for inverse-CNR weighting
(continued)

Metric	Training Only Unweighted	Training Only Inv.-CNR	Training + Testing Unweighted	Training + Testing Inv.-CNR
Final SINR (dB)	8.2492	18.6234	11.9746	13.0376
Train time (s)	148.755	95.509	117.507	97.422
Test time (s)	0.000	0.000	1.976	2.110
Total time (s)	148.755	95.509	119.483	99.531